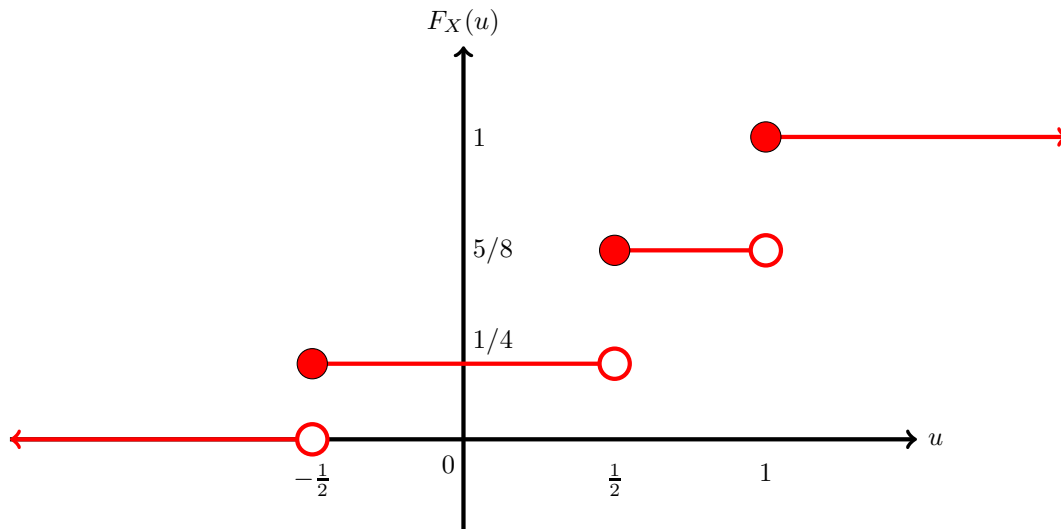


The cumulative distribution function of a random variable  $X$  is shown below. What is the probability that  $-\frac{1}{2} \leq X \leq 1$  ?



$$P(-\frac{1}{2} \leq X \leq 1) = 1$$

- (a) None of these
- (b)  $3/16$
- (c)  $1/2$
- (d)  $1/4$
- (e)  $5/8$
- (f)  $3/8$
- (g)  $3/4$
- (h)  $2/3$
- (i)  $1/8$
- (j)  $7/8$
- (k)  $0$
- (l)  $1/16$

An experiment has sample space  $S = \{a, b, c, d, e\}$  with probabilities  $P(a) = P(c) = 1/8$  and  $P(b) = P(d) = P(e) = 1/4$ . Define random variables  $X$  and  $Y$  on  $S$  as follows:

$$\begin{aligned} X(a) &= 0 & Y(a) &= 1/8 \\ X(b) &= 1/2 & Y(b) &= 1/16 \\ X(c) &= 1/4 & Y(c) &= 0 \\ X(d) &= 1/4 & Y(d) &= 1/4 \\ X(e) &= 1/8 & Y(e) &= 1/8. \end{aligned}$$

What is the probability that  $X = Y$  or  $X < 0.2$  (or both) ?

(a)  $5/8$

(b)  $1/16$

(c)  $1/2$

(d)  $1/4$

(e)  $3/8$

(f)  $3/4$

(g)  $2/3$

(h)  $1/8$

(i)  $7/8$

(j)  $0$

(k)  $0.2$

(l) None of these

$$\begin{aligned} P(X=Y \cup X < 0.2) &= P(a) + P(d) + P(e) \\ &= \frac{1}{8} + \frac{1}{4} + \frac{1}{4} = \left(\frac{5}{8}\right) \end{aligned}$$

a ✓

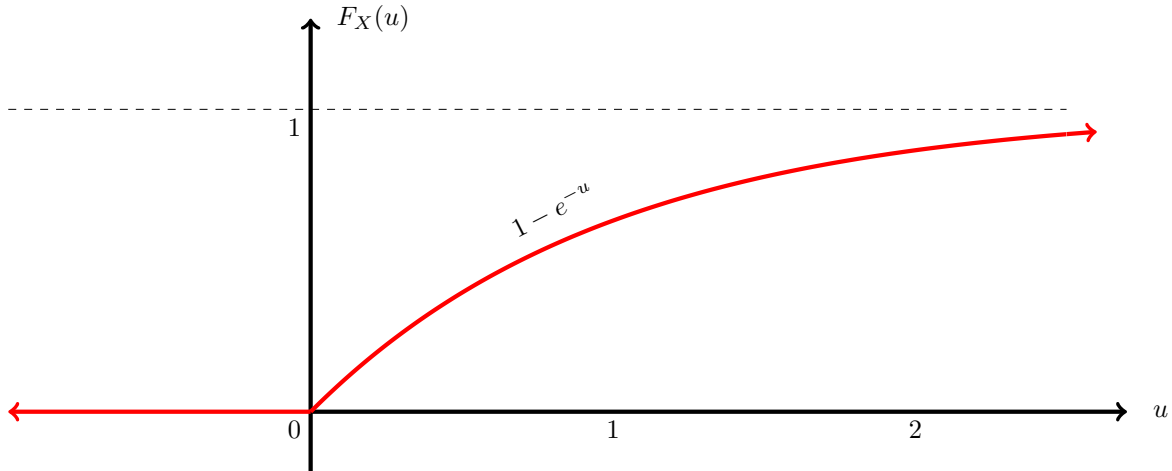
b ✗

c ✗

d ✓

e ✓

Let  $X$  be a random variable whose cumulative distribution function is shown below. What is the probability that  $e^X$  is greater than 1.5 ?



$$P(e^X > 1.5)$$

$$= P(X > \ln(1.5))$$

$$e^{-\ln(x)} = e^{\ln(x^{-1})} = \frac{1}{x}$$

(a)  $2/3$

(b)  $1/3$

(c)  $1/e$

(d)  $(e-1)/e$

(e)  $1/2$

(f)  $3/4$

(g)  $1/4$

(h)  $e/3$

(i)  $e/2$

(j)  $1/\ln 2$

(k)  $1/\ln 3$

(l) None of these

$$P(X > u) = 1 - P(X \leq u)$$

$$= 1 - F_X(\ln(1.5))$$

$$= 1 - (1 - e^{-\ln(1.5)})$$

$$= 1 - (1 - e^{-\ln(\frac{3}{2})})$$

$$= 1 - (1 - \frac{2}{3})$$

$$= \frac{2}{3}$$